Booleans and Comparisons

Truth values, comparisons, and explicit conversions

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Breaking Changes

New keywords: `true`, `false`, `boolean`, `type`

Rename earlier variables that use those exact names.

`true_value`, `boolean_value`, and `typewriter` remain identifiers.

Computing an Answer

```
__input = "21";
answer = number(input()) * 2;
large_enough = answer >= 40;
print(large_enough);
```

```
true
```

A New Literal Node

```
ready = true;
```

```
{"tag": "boolean", "value": True}
```

Vertex spelling: `true`. Python value: `True`.

Comparing Values

Expression	Result
`42 == 42.0`	`true`
`"dog" != "cat"`	`true`
`-1.5 < .5`	`true`
`2 <= 2`	`true`
`2 > 3`	`false`
`3 >= 3`	`true`

Ordering Strings

Expression	Result
`"ant" < "bee"`	`true`
`"ant" < "anteater"`	`true`
`"ant" <= "ant"`	`true`
`"Z" < "a"`	`true`
`"10" < "2"`	`true`

Compare characters, not numeric meanings.

Types Still Matter

Expression	Result
`true == 1`	`false`
`"1" == 1`	`false`
`"1" < 2`	Type error
`true + 1`	Type error
`number(true) + 2`	`3`

Explicit Conversions

Expression	Value
`number(true)`	`1`
`string(false)`	`"false"`
`boolean(-2)`	`true`
`boolean(0)`	`false`
`boolean(" FALSE ")`	`false`

Text Needs an Interpretation

```
boolean("false")          false
boolean("1")              value error
boolean(number("1"))      true
```

No nonempty-string truthiness.

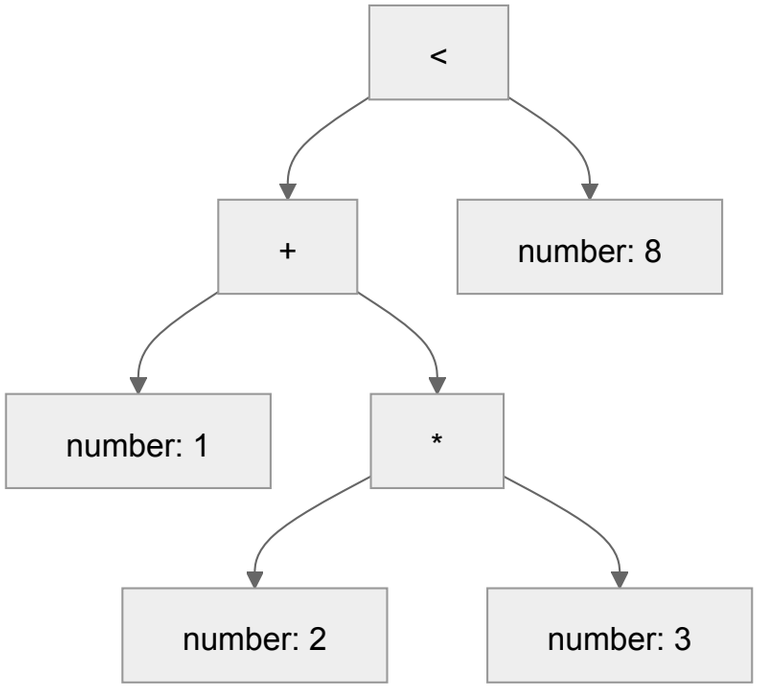
The Type of a Value

Expression	Result
`type(42)`	`"number"`
`type(.5)`	`"number"`
`type("42")`	`"string"`
`type(1 < 2)`	`"boolean"`

`type(input())` still reads input.

The Comparison Tree

```
1 + 2 * 3 < 8
```



One Comparison at a Time

```
expression ::= comparison
comparison ::= arithmetic_expression
               [ compare_op arithmetic_expression ]
```

Expression	Result
`1 < 2 < 3`	Syntax error
`(1 < 2) == true`	`true`
`(1 < 2) < 3`	Type error

Checking the Implementation

```
bash run-tests.sh
```

All six comparisons. Exact result types. Conversion boundaries.

Input arguments run once; comparison operands run left to right.

Practice

```
true == 1
number(true) == 1
boolean("0")
boolean(number("0"))
```

Which succeed, and what does each produce?